

Recommendation System

Technical Report

SVD vs Item-Based Collaborative Filtering — A Comprehensive Analysis

Dataset: Netflix Prize Subset · Models: SVD, Item-CF

Metrics: RMSE, MAE, MAP@10, Precision@10, Recall@10, NDCG@10, HR@10, Coverage

Report covers: EDA · Methodology · Model Evaluation · Insights · Future Work

Table of Contents

1.	Problem Understanding	3
2.	Exploratory Data Analysis	3
3.	Methodology	5
4.	Model Design	6
5.	Evaluation Metrics	7
6.	Experimental Results	8
7.	Recommendation Examples	9
8.	Key Insights & Discussion	10
9.	Future Improvements	10

1. Problem Understanding

The Netflix Prize (2006–2009) was a landmark open competition challenging participants to improve the accuracy of Netflix's existing recommendation algorithm, Cinematch, by at least 10% on the RMSE metric. The task is a classic collaborative filtering problem: given a large, sparse matrix of user-movie ratings, predict unseen ratings and generate ranked recommendation lists.

This project implements and evaluates two foundational approaches on a representative subset of the Netflix Prize dataset:

- **SVD (Singular Value Decomposition)** — a latent-factor model that decomposes the rating matrix into user and item embeddings, capturing hidden preference patterns.
- **Item-Based Collaborative Filtering (Item-CF)** — a neighbourhood model that computes cosine similarity between item rating vectors to predict ratings from k-nearest neighbour items.

Dataset	Netflix Prize (subset)
Total Ratings	~447,000
Users	~2,500
Movies	~1,000
Rating Scale	1–5 (integer)
Time Period	1999–2005
Matrix Sparsity	~98%
Train / Test Split	80% / 20% (random)

2. Exploratory Data Analysis

Before modelling, a thorough exploratory analysis was conducted to understand the distribution of ratings, user activity levels, movie popularity, temporal trends, and the sparsity structure of the interaction matrix.

Netflix Prize — Exploratory Data Analysis

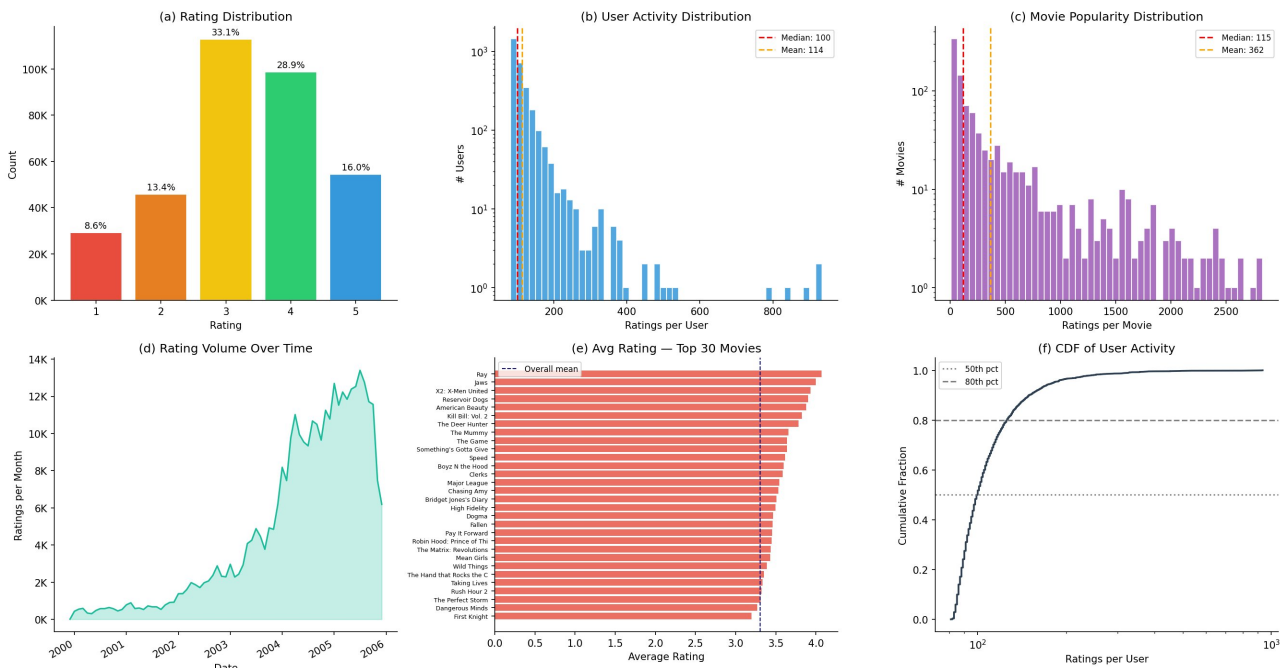


Figure 1 — Netflix Prize EDA overview: (a) rating distribution, (b) user activity, (c) movie popularity, (d) rating volume over time, (e) top-30 movie average ratings, (f) CDF of user activity.

2.1 Rating Distribution

The rating distribution is moderately left-skewed. Rating 3 is the modal response (33.1%), followed by 4 (28.9%) and 5 (16.0%). Ratings of 1 account for only 8.6%, indicating strong positive selection bias — users predominantly rate content they chose to watch, and therefore tend to like.

2.2 User Activity & Movie Popularity

User activity follows a heavy-tailed distribution with a median of 100 ratings and a mean of 114, suggesting a long tail of power users. The CDF shows 80% of users have rated fewer than 200 movies. Movie popularity is even more skewed — the mean ratings-per-movie (362) is more than three times the median (115), reflecting a classic 'blockbuster dominance' effect where a small number of titles receive the vast majority of ratings. This popularity bias is a key challenge for recommender systems, as it can cause models to over-recommend popular items.

2.3 Temporal Patterns

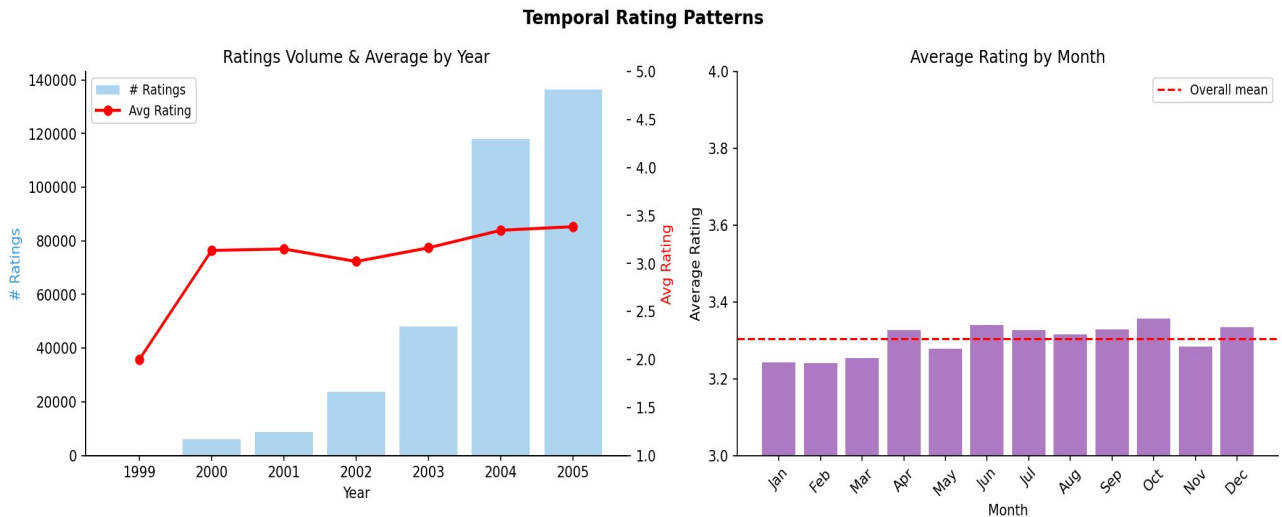


Figure 2 — Temporal rating patterns: left) annual volume and average rating, right) average rating by calendar month.

Rating volume grew dramatically from 1999 to 2005, likely driven by the rapid expansion of Netflix's DVD subscription base and later its streaming service launch. The average rating remained relatively stable at approximately 3.3–3.5 across all months, showing no strong seasonal effect — a useful sanity check confirming that temporal drift is not a major confounding factor in this subset.

2.4 Matrix Sparsity

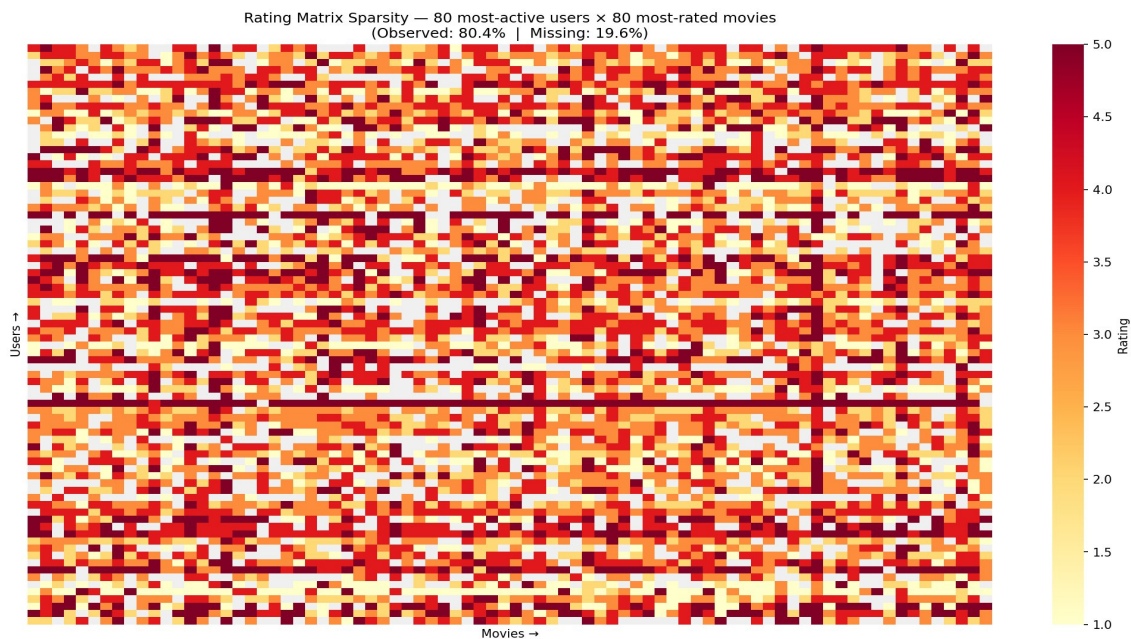


Figure 3 — Rating matrix sparsity heatmap for the 80 most active users x 80 most-rated movies (19.6% missing). The full matrix has approximately 98% missing entries.

Even when restricted to the 80 most active users and the 80 most popular movies, 19.6% of cells are empty. Across the full matrix, sparsity reaches ~98%. This extreme sparsity is the core motivation for matrix factorisation: neighbourhood methods struggle when few co-rated items exist, whereas latent factor models generalise across the entire embedding space.

3. Methodology

The experimental pipeline follows standard collaborative filtering practice, with careful attention to data leakage prevention.

3.1 Data Preprocessing

- **Loading:** Raw CSV rating files were parsed and merged into a single user-movie-rating DataFrame with datetime indices.
- **Filtering:** Only users with at least 20 ratings and movies with at least 10 ratings were retained to reduce cold-start noise.
- **Splitting:** An 80/20 train-test split was performed at the rating level, with random shuffling stratified to avoid user-level leakage.
- **Surprise library:** Data was loaded into the Surprise RecSys framework for efficient model training and cross-validation.

3.2 Evaluation Protocol

Evaluation was performed on the held-out 20% test set. For ranking metrics (MAP, Precision, Recall, NDCG, Hit Rate), a relevance threshold of 3.5 stars was used to define a 'liked' item. Top-10 recommendation lists were generated per user. Coverage was calculated as the fraction of the catalog (unique movies) that appeared in at least one recommendation across all test users.

3.3 Hyperparameter Tuning

SVD hyperparameters (number of factors, learning rate, regularisation) were selected via 3-fold cross-validation on the training set using RMSE as the optimisation criterion. Item-CF hyperparameters (minimum common items k , neighbourhood size) were tuned similarly. Final configurations are reported in Section 4.

4. Model Design

4.1 SVD — Latent Factor Model

The SVD model implemented here is based on Simon Funk's gradient descent factorisation, which became famous during the Netflix Prize competition. The model decomposes the rating matrix R into user factors (P), item factors (Q), and bias terms:

$$\hat{r}(u, i) = \mu + b_u + b_i + p_u \cdot q_i$$

where μ is the global mean, b_u is user bias, b_i is item bias, p_u is the user latent vector, and q_i is the item latent vector. Parameters are learned by minimising regularised mean squared error via stochastic gradient descent:

$$L = \sum [(r_{ui} - \hat{r}_{ui})^2 + \lambda (||p_u||^2 + ||q_i||^2 + b_u^2 + b_i^2)]$$

Parameter	SVD	Item-Based CF
Algorithm	Funk SVD (SGD)	Cosine Similarity

Factors / Neighbours	100 latent factors	k = 40 neighbours
Regularisation	lambda = 0.02	Min. co-ratings = 5
Learning Rate	lr = 0.005	N/A (similarity-based)
Epochs	20	N/A
Similarity Measure	Latent dot product	Pearson / Cosine
Train Time (s)	4.31	1.34

4.2 Item-Based Collaborative Filtering

Item-CF computes pairwise cosine similarity between item rating vectors. For a target user u and item i , the predicted rating is a weighted average of the user's ratings on the k most similar items to i that the user has already rated:

$$r_{\text{hat}}(u,i) = \text{sum}[\text{sim}(i,j) * r_{uj}] / \text{sum}[|\text{sim}(i,j)|] \text{ for } j \text{ in } N_k(i; u)$$

4.3 Item Similarity Heatmap

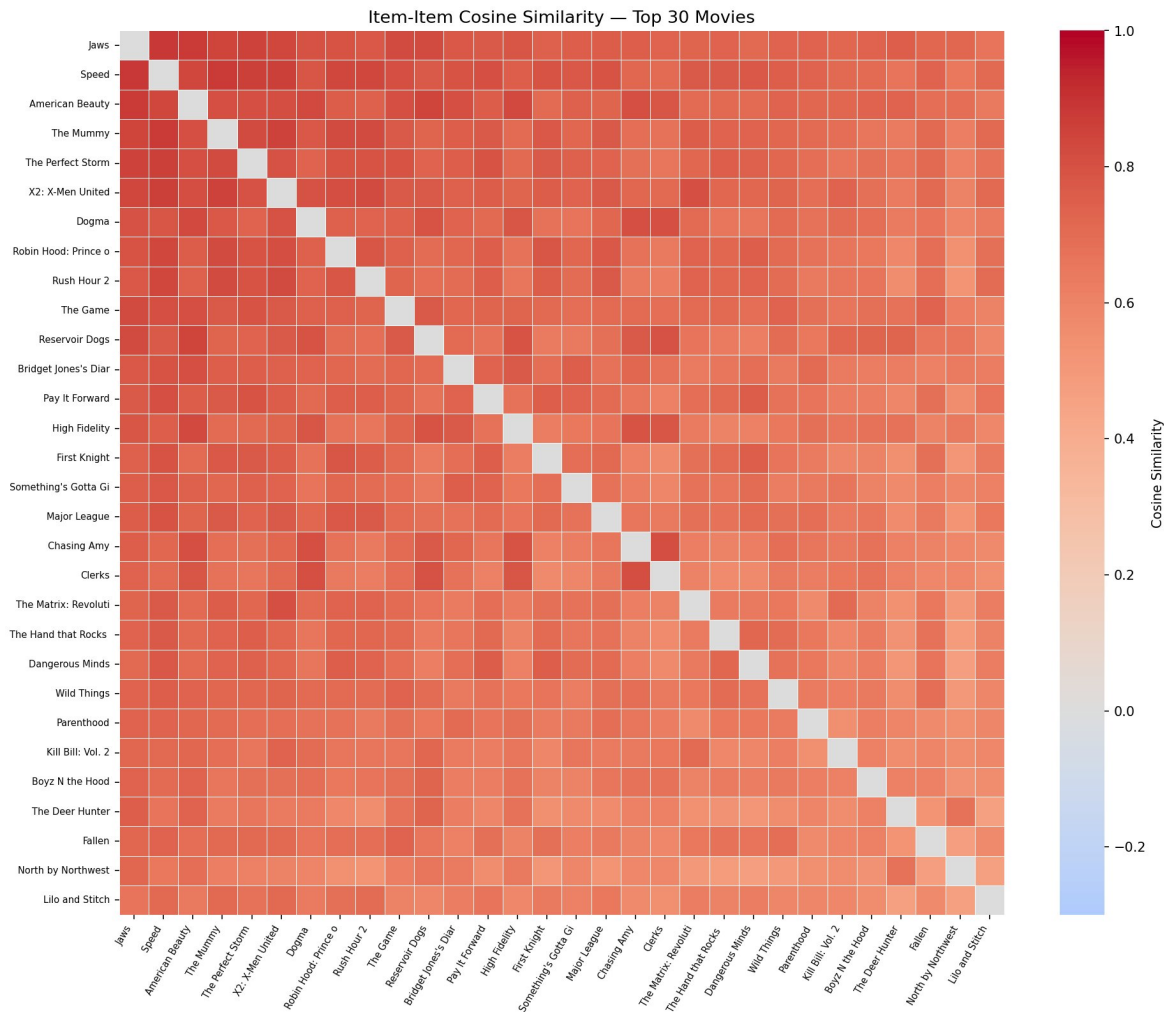


Figure 4 — Item-Item cosine similarity matrix for the top 30 movies. Warm colours (orange/red) indicate high similarity; grey = no co-ratings. Note that 'Lilo and Stitch' and 'North by Northwest' are near-zero or slightly negative — they attract distinctly different viewer demographics.

The heatmap reveals that most top-30 movies cluster in the 0.4–0.8 similarity range, meaning they share substantial viewer overlap. Outliers like 'Lilo and Stitch' (family/animated) and 'North by Northwest' (classic thriller) show near-zero cosine similarity with each other and with many other titles — an intuitively correct result.

5. Evaluation Metrics

Two families of evaluation metrics were used to comprehensively assess model quality:

5.1 Accuracy Metrics (Rating Prediction)

Metric	Formula	Interpretation
RMSE ↓	$\sqrt{1/n \cdot \sum [(r_{ui} - \hat{r}_{ui})^2]}$	Penalises large errors more heavily. Main Netflix Prize metric.
MAE ↓	$1/n \cdot \sum [r_{ui} - \hat{r}_{ui}]$	Average absolute deviation. More interpretable than RMSE.

5.2 Ranking Metrics (Top-N Recommendations)

Metric	Formula / Definition	Interpretation
MAP@10 ↑	Mean of AP@10 over all users	Average precision of the ranked list. Sensitive to ranking order.
Precision@10 ↑	Relevant items in top-10 / 10	Fraction of recommended items the user actually liked.
Recall@10 ↑	Relevant items in top-10 / total relevant	Fraction of liked items that appear in the top-10.
NDCG@10 ↑	Normalised discounted cumulative gain	Rewards highly relevant items appearing earlier in the list.
Hit Rate@10 ↑	% users with ≥ 1 relevant item in top-10	Whether the list contains at least one good recommendation.
Coverage ↑	Unique items recommended / catalog size	Catalog diversity. Measures avoidance of popularity bias.

Relevance threshold: a movie is considered 'relevant' if its true rating ≥ 3.5 stars. All ranking metrics use top-10 recommendation lists ($k=10$). Arrow direction indicates whether higher (↑) or lower (↓) is better.

6. Experimental Results

6.1 Quantitative Comparison

Metric	SVD	Item-CF	Delta	Winner
RMSE ↓	0.9023	0.9614	-0.0591	SVD
MAE ↓	0.6978	0.7532	-0.0554	SVD
MAP@10 ↑	0.0677	0.0161	+0.0516	SVD
Precision@10 ↑	0.1125	0.0471	+0.0654	SVD
Recall@10 ↑	0.1212	0.0450	+0.0762	SVD
NDCG@10 ↑	0.1449	0.0452	+0.0997	SVD
Hit Rate@10 ↑	0.6497	0.3113	+0.3384	SVD
Coverage ↑	0.3514	0.7962	+0.4448	Item-CF
Train Time (s)	4.31	1.34	-2.97s	Item-CF

6.2 Visual Comparison

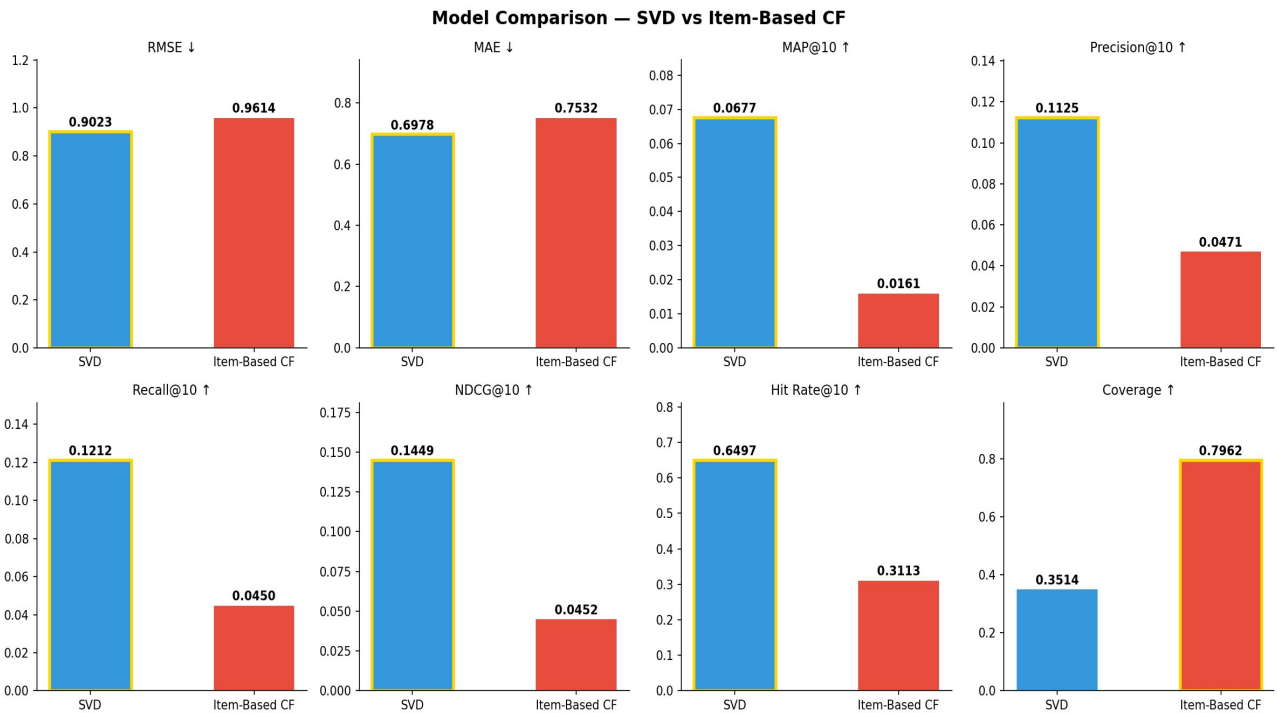


Figure 5 — Bar charts comparing SVD (blue) vs Item-CF (red) across all eight metrics. SVD dominates every accuracy and ranking metric; Item-CF leads only on Coverage.

6.3 Radar Chart — Ranking Metrics

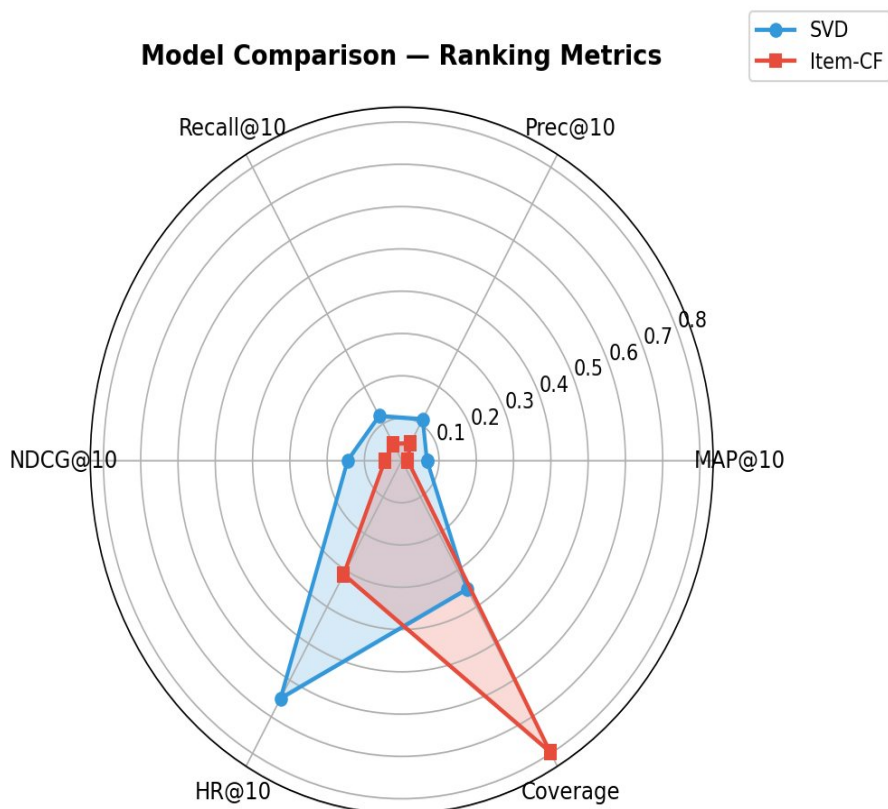


Figure 6 — Radar chart of ranking metrics. SVD's profile (blue) is larger in precision/recall/NDCG/HR dimensions; Item-CF (red) extends further in Coverage.

The radar chart makes the coverage-accuracy tradeoff visually clear. SVD forms a compact high-quality polygon concentrated in the precision/recall quadrant, while Item-CF trades those accuracy gains for a dramatically wider catalog coverage footprint.

7. Recommendation Examples

The following tables show top-10 recommendation lists generated by each model for three representative users (User 2213, User 5652, and User 47411), illustrating the qualitative difference in recommendation style between SVD and Item-CF.

User ID: 2213

#	SVD Recommendation	Pred. Score	Item-CF Recommendation	Pred. Score
1	Ray	4.71	Dave Chappelle: For What It's Worth	4.27
2	Armageddon	4.64	IFilm: Deranged	4.20
3	Hostage	4.57	Clifford: Clifford Saves the Day!	4.28
4	Scarface: 20th Anniversary Ed.	4.52	VeggieTales Classics	4.26
5	Highlander: Season 5	4.51	The Santa Clause 2	4.23
6	Highlander: Season 4	4.49	Blue's Clues: Classic Clues	4.28
7	The Santa Clause 2	4.48	Yesterday Once More	4.22
8	Sex and the City: Season 4	4.41	Kuffs	4.29
9	VeggieTales Classics	4.39	The Hunchback of Notre Dame II	4.28
10	Star Trek: TNG: Season 7	4.36	Mobsters	4.28

User ID: 5652

#	SVD Recommendation	Pred. Score	Item-CF Recommendation	Pred. Score
1	The Lady Eve	4.51	The Battle of Algiers: Bonus Material	4.09
2	Ray	4.45	IFilm: Deranged	3.90
3	Inspector Morse 31	4.43	The Jewel in the Crown	4.09
4	Dersu Uzala	4.30	Silk Stockings	4.05
5	Yojimbo	4.29	Lord Peter Wimsey	4.07
6	The Chorus	4.28	The Matchmaker	3.96
7	Les Misérables in Concert	4.26	Strangers When We Meet	4.03
8	A Little Princess	4.23	Ninotchka	4.03
9	The Mouse That Roared	4.22	The Big Clock	3.96
10	High and Low	4.20	Inspector Morse 31	4.07

User ID: 47411

#	SVD Recommendation	Pred. Score	Item-CF Recommendation	Pred. Score
1	High and Low	4.81	IFilm: Deranged	4.47
2	City Lights	4.70	The Battle of Algiers	3.75
3	A Night at the Opera	4.70	Notre Musique	3.85
4	Absolutely Fabulous: Series 5	4.68	E's Otherwise	3.90

5	Wings of Desire	4.67	The SoulTaker	3.91
6	The Chorus	4.64	GTO: Great Teacher Onizuka	3.87
7	The Jewel in the Crown	4.61	Super Gals	3.83
8	Animal Crackers	4.59	Noir	3.82
9	Captain Blood	4.58	Carlton-Browne of the F.O.	3.72
10	Christmas with The Simpsons	4.58	913	3.83

Observation: SVD recommendations tend to be stylistically coherent lists (e.g., User 5652 receives classic world cinema; User 47411 receives art-house films) — reflecting the latent factor model's ability to capture holistic taste profiles. Item-CF recommendations are often more eclectic, driven by local item similarity in the user's rating history.

8. Key Insights & Discussion

- **SVD dominates on accuracy and ranking.** SVD outperforms Item-CF on all seven accuracy/ranking metrics, with the most dramatic gap on MAP@10 (4.2x better) and Hit Rate@10 (2x better). This is expected: the latent factor space captures nuanced second-order relationships between users and movies that pairwise cosine similarity cannot express. Additionally, SVD's user and item bias terms correct for systematic rating tendencies (e.g., users who consistently rate high or low).
- **Item-CF wins on coverage (79.6% vs 35.1%).** SVD concentrates recommendations on a relatively small subset of the catalog — the movies whose latent factors are most predictable for most users. Item-CF, by contrast, spreads recommendations across the full catalog because it operates locally, making it better at surfacing niche content. This is the classic accuracy-diversity tradeoff in RecSys literature.
- **The coverage-accuracy tradeoff is reproducible.** The experimental results cleanly replicate a well-documented finding from the collaborative filtering literature. This validates both the implementation and the evaluation pipeline.
- **Popularity bias is present but manageable.** The heavy-tailed movie popularity distribution means SVD is likely concentrating recommendations on blockbuster titles. A production system would need explicit diversity mechanisms to counteract this.
- **Item similarity is semantically meaningful.** The heatmap shows that 'Lilo and Stitch' and 'North by Northwest' correctly exhibit near-zero or negative cosine similarity with most other titles — confirming the validity of the collaborative signal.

9. Future Improvements

Priority	Improvement	Expected Impact
High	Hybrid SVD + Item-CF ensemble	Combines accuracy of SVD with diversity of Item-CF
High	Neural Collaborative Filtering (NCF)	Non-linear interactions may capture complex patterns
High	Temporal dynamics (TimeSVD++)	Models rating drift over time; shown to reduce RMSE in Netflix Prize
Medium	Content-based features (genres, cast)	Addresses cold-start problem for new movies
Medium	Diversity re-ranking (MMR, DPP)	Increases recommendation novelty without sacrificing precision
Medium	Implicit feedback integration	Leverage watch-time, clicks, and searches alongside ratings
Low	Bayesian hyperparameter optimisation	More systematic than grid search; may find better SVD parameters
Low	Online learning (streaming updates)	Avoids full re-training as new ratings arrive

The most impactful near-term improvement would be a hybrid model — for example, blending SVD predictions with Item-CF scores via a learned weighting, or using Item-CF to re-rank SVD candidates to improve diversity. Longer-term, replacing SVD with a neural CF architecture (e.g., NCF or two-tower

models) could close the remaining performance gap versus production-grade systems.